

Course: Artificial Intelligence
Course code: CSA 1709

Name: E. Vidhya Sagar
Regno: 192565024

Answer the following:

Smart home Automation system and Intelligent agent

Introduction:

A smart home automation system is a special mode of agent AI for home purpose. Such as this system controls lighting, Temperature and security, using sensors such as motions, detectors and clocks. Artificial intelligence makes the system intelligent and also efficient in decision making based on the environment conditions.

Types of intelligent agents:

1. Simple-reflex agent
2. Model-based agent
3. Goal-based agent
4. Utility-based agent

Simple-reflex agent:

- * Detection sensors are used in this case
- * It does not have any memory for this agent
- * If the agent detects something the action will be

Example: If the Person comes in to the room the light gets turn on and he went off light gets turn off (Detection system)

ii, Model-Based agent:

A Model Based agent remembers an Previous Inform and keeps an internal model of the environment for the action execution.

Example: -> Living room is occupied, Bedroom is empty etc
It uses memory and Better decision making.

iii, Goal-Based agent:

Goal Based agent makes decisions based on achieving goal

Example: Reduce electricity usage

It also have goal-oriented tasks and Better Planning

Utility-Based agent:

A utility-Based agent chooses the action that gives the best overall agent

It consider many factors before running and making a decision

Eg: The system checks user comfort, security and electricity cost

Suppose

Temperature = 29°C

Instead of setting AC to 18°C , it sets it to 24°C

Comparison table:

Agent type	memory	goal	learns	Example
Simple Reflex	NO	NO	NO	Motion detection
Model-Based	Yes	NO	Partial	Remember occupied rooms
Goal-Based	Yes	Yes	NO	Goal oriented
Utility Based	Yes	Yes	Yes	chairs are comfortable

2. Robot in an unknown maze using a search algorithm

Introduction:

A robot is placed in an unknown maze and must find the exit without knowing the map. Artificial intelligence uses a searching algorithm to explore a maze and determine a Best Path.

Two common uninformed search algorithms

- i, Uniform cost search (UCS)
- ii, Iterative Deepening search (IDS)

An uninformed is a search method also called as a blind search

Iterative Deepening search:

IDS repeatedly performs a Depth first search with increasing with a depth limits

1. Search with depth = 0
2. If not found increase depth to 1
3. Continue until exit is found

Best combination:

Broad-Based agent uniform cost search

Justification:

Broad Based agent always focused on reaching an goal which is exit from the a maze. It can guarantee the determination of an best path to exit.